ECE 30100 - Signals and Systems, Section 3 (CRN 17101)

Course Information
Fall 2025

Instructor: Professor Kwang Taik Kim
Room: MSEE 340

Graduate Teaching Assistant: Jiahui Ni and Jesse Wayment

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Course Objectives: A student who successfully fulfills the course requirements will have demonstrated:

- an ability to classify signals (e.g. periodic, even) and systems (e.g. causal, linear) and an understanding of the difference between discrete and continuous time signals and systems.
- an ability to determine the impulse response of a differential or difference equation.
- an ability to determine the response of linear systems to any input signal by convolution in the time domain.
- an understanding of the definitions and basic properties (e.g., time-shift, modulation, Parseval's Theorem) of Fourier series, Fourier transforms, Z transforms, and discrete time Fourier transforms and an ability to compute the transforms and inverse transforms of basic examples using methods such as partial fraction expansions.
- an ability to determine the response of linear systems to any input signal by transformation to the frequency domain, multiplication, and inverse transformation to the time domain.
- an ability to apply the sampling theorem, reconstruction, aliasing, and Nyquist's theorem to represent continuous-time signals in discrete time so that they can be processed by digital computers.

Course Descriptions: The official course description from the university catalog can be found at https://engineering.purdue.edu/ECE/Academics/Undergraduates/UGO/CourseInfo/courseInfo?courseid=33&show=true&type=undergrad

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Required text:

- *Signals and Systems*, 2nd Edition, Oppenheim, Willsky, and Nawab, Prentice-Hall, 1996, ISBN No. 9780138147570.

Pre-requisites: (ECE 20002 Minimum Grade of C or ECE 20200 Minimum Grade of C or BME 30500) and (MA 26200 or MA 36600 or MA 26600).

Grading: Each student's grade is determined using two distinct methods, and the higher of the two grades is selected:

- Calculation method A (diligent type)
  - Homework (10%), Quiz (5%), Midterm 1 (15%), Midterm 2 (15%), Midterm 3 (15%), Final (40%).
- Calculation method B (late blooming type)
  - Homework (10%), Quiz (5%), Midterm 1 (5%), Midterm 2 (15%), Midterm 3 (25%), Final (40%).

NOTE: This section is being taught completely independently of the other sections of this course offered this semester. All assessments for this section are different, and grades are assigned with an independent grading scale for this section relative to the other sections.

Quizzes: There will be a short 10-minute closed-book quiz mid-class at scattered times during the semester. I almost always announce the quizzes in advance, but there may be surprises. The quiz questions will focus on the assigned reading, on basic knowledge, and individual steps of the multi-step process required to solve problems. As such, they may contain material that you have not yet seen on homeworks! The lowest 2 quiz scores will be dropped.

Evening Exams: There will be three midterms and a final. All exams will be closed-book closed-notes with **no calculators and no cheat sheets**. The final exam will be cumulative and take place during finals week. One lecture will be canceled to compensate for each evening exam.

Homeworks: Homeworks will be assigned every week or two, and due Wednesdays at 11:59pm. This semester we will use Gradescope for all homework problems. Gradescope requires you to electronically submit homeworks, and enables us to provide feedback on them. **Late homework will not be accepted,** but as a courtesy, Gradescope will remain open for submissions until 6:59am the following morning.

At the top of each homework, on PAGE ONE, you **MUST** briefly list all sources of information you used, not counting the text, handouts, or material posted on the web page. Examples:

- "Problems 3,4 and 5 I did with students X and Y."
- "I used article AB to solve this problem."
- "I looked at the solutions from prior class Z on the web to solve this problem."

If you do not use any additional sources other than the text, handouts, or material posted on the web page for this course for this semester, you **MUST** state at the top of the homework "I completed this homework on my own without external help". Homework without any such statement at the top will not be accepted.

Note that there exists homework "help" online to some problems. BEWARE! Not everything you read online is true. If you copy an online homework solution that is known to be wrong, and if you have stated that you completed the homework on your own without external help, then you will receive no credit for your homework. This will also be considered cheating, and will be reported to the Dean of Students.